Direct Tangent-Actor value head across dimensions

Whole-function value RMSE on matched frozen state banks · lower is better

PDE	d	Direct value head	DPI	PINN-HTE	SDGD-PINN	Head wins
Allen–Cahn	20	2.55e-3	**1.45e-3**	1.72e-3	1.19e-2	1/3
	50	1.98e-3	1.44e-3	**1.31e-3**	6.12e-3	1/3
	100	1.37e-3	1.07e-3	**9.48e-4**	2.69e-3	1/3
	200	8.53e-4	9.04e-4	**8.47e-4**	1.42e-3	2/3
	400	5.86e-4	4.55e-4	**4.26e-4**	6.16e-4	1/3
Concave Cubic–Quadratic	20	9.86e-4	**8.81e-4**	3.83e-3	2.59e-2	2/3
	50	1.18e-3	**9.67e-4**	2.80e-3	1.80e-2	2/3
	100	2.36e-3	**2.14e-3**	3.23e-3	2.33e-2	2/3
	200	3.11e-3	**2.76e-3**	3.43e-3	2.79e-2	2/3
	400	5.29e-3	6.82e-3	**5.26e-3**	2.97e-2	2/3
Convex Cubic–Quadratic	20	**5.79e-4**	7.28e-4	1.62e-3	1.33e-2	3/3
	50	1.23e-3	**9.71e-4**	2.68e-3	1.50e-2	2/3
	100	1.97e-3	**1.86e-3**	3.27e-3	1.28e-2	2/3
	200	4.13e-3	**2.93e-3**	4.60e-3	1.68e-2	2/3
	400	7.59e-3	**4.09e-3**	5.06e-3	1.38e-2	1/3
Oscillatory Value–Gradient	20	2.16e-3	**1.08e-3**	7.09e-3	1.27e-2	2/3
	50	3.42e-3	**1.08e-3**	4.27e-3	1.37e-2	2/3
	100	4.72e-3	**1.70e-3**	3.07e-3	8.88e-3	1/3
	200	9.23e-3	5.22e-3	**3.67e-3**	1.04e-2	1/3
	400	2.37e-2	6.15e-3	**4.93e-3**	1.33e-2	0/3
Fourier–Tanh Coupled	20	**7.12e-3**	2.06e-2	4.90e-2	1.35e-2	3/3
	50	**1.47e-2**	6.25e-2	6.55e-2	5.13e-2	3/3
	100	**1.18e-2**	6.46e-2	7.71e-2	7.41e-2	3/3
	200	**4.29e-2**	2.46e-1	2.62e-1	2.09e-1	3/3
	400	**5.98e-2**	3.90e-1	4.01e-1	2.33e-1	3/3

Pairwise wins: 47/75 (62.7%) · vs DPI 8/25 · vs PINN-HTE 15/25 · vs SDGD-PINN 24/25

Bold = lowest RMSE in row